

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	Assessment:	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	Total Marks:	100
CO5 Statement:	<i>Construct MST using shortest path algorithms and traverse the graph to model and analyse realworld problem.</i>		
Student Name:	KOPPALA NAVYA	Reg. No.:	192511231
Section / Batch:	B.E CSE	Date:	28/08/2026

Code of Conduct

I ___KOPPALA NAVYA(192511231)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: KOPPALA NAVYA

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20

Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or nonfunctional code
Code Quality & Readability	20	Clean, modular, wellcommented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

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QUESTIONS

Q1
Minimum Spanning Tree - Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - MST & Dijkstra's Algorithm
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST. (20 Marks)
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

SECOND_BEST_MST(G):

// Step 1: Find the Minimum Spanning Tree
MST = KRUSKAL(G)

if MST does not contain V - 1 edges:

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QUESTIONS

Q1
Minimum Spanning Tree - Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - MST & Dijkstra's Algorithm
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q1 Minimum Spanning Tree - Kruskal's Algorithm20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).
Constraints:
a. All zones must be connected
b. Total installation cost must be minimized
c. No redundant connections (no cycles) allowed
d. The network must remain fully connected

Your Answer

This problem describes finding a Minimum Spanning Tree (MST) of the graph.

Why MST fits

All zones must be connected → every vertex must be reachable.

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QUESTIONS

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Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q2 Graph - Topological Sort20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.
Constraints:
a. Detect cycle before sorting
b. Use DFS approach

Your Answer

Topological Sort using DFS with Cycle Detection

TOPOLOGICAL_SORT(G):
 visited[v] = false for every vertex v
 inStack[v] = false for every vertex v
 stack = empty

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QUESTIONS

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Q2
Graph - Topological Sort
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Q3
Shortest Path Algorithm - MST
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20 marks

Q4
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q3 Shortest Path Algorithm - MST & Dijkstra's Algorithm20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.
Constraints:
a. Use MST for infrastructure
b. Use shortest path for routing

Your Answer

For a smart transportation system, use Kruskal's algorithm to build the minimum-cost road infrastructure (MST), then use Dijkstra's algorithm on that MST to find the shortest route.

SMART_TRANSPORTATION(G, source, destination):

 // Step 1: Build MST using Kruskal's Algorithm

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QUESTIONS

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Minimum Spanning Tree -
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20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - MST
& Dijkstra's Algorithm
20 marks

Q4
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q4 Shortest Path Algorithm - Dijkstra's Algorithm20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

DIJKSTRA_WITH_VOUCHER(G, S, D):

for each vertex v:
 dist[v][0] = ∞
 dist[v][1] = ∞

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